Attention as Quantum State: The Gibbs State Construction and Exact Born Rule Correspondence

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Abstract

We provide the explicit physical construction that rigorous connection between transformer attention and quantum measurement theory requires. We show that the attention mechanism is the classical limit of a natural quantum system: the attention weights are the diagonal elements of a Gibbs state density matrix in a finite-dimensional key Hilbert space, and the attention output is the quantum expectation value of the value observable in this state. The diagonal (classical) construction is exact but not specific to attention — any probability distribution embeds as a diagonal density matrix. What *is* specific: (a) the Gibbs state construction gives a natural and well-defined framework for quantum extension via off-diagonal coherences; (b) the classical Fisher-Rao metric that Kim (2026) uses to characterize attention geometry is *identical* to the quantum Fisher information metric for the diagonal Gibbs state (Theorem 4), closing Junction 2 as an exact mathematical identity; and (c) the off-diagonal extension defines a genuinely new class of "quantum attention" mechanisms with coherence between key positions. We derive: (1) the exact Gibbs state construction; (2) the Born rule as an exact theorem for diagonal states; (3) closure of Junction 2; (4) the Kraus operator representation of the attention channel; and (5) the quantum extension to off-diagonal density matrices. Together with Paper 4's SYK construction, these results ground the holographic attention framework in explicit physical constructions.

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1. Introduction: The Explicit Construction Question

In response to expert critique of Papers 1–3 that the connections to quantum measurement theory required "a much more explicit physical construction," Paper 4 addressed the holographic junction (Junctions 3–4) through the Sachdev-Ye-Kitaev model. The quantum measurement connection (Junction 2) remained open. This paper closes it.

The claim of Papers 2 and 3 was that transformer attention implements quantum measurement. The critique was that this resemblance — however mathematically suggestive — does not constitute a physical equivalence: two systems can have similar equations without being the same system in the relevant physical sense.

We show here that the attention mechanism defines a quantum density matrix (the Gibbs state) whose diagonal elements are the attention weights, and computes a quantum expectation value of the value observable. The diagonal Gibbs state is classical — any probability distribution embeds this way. What makes the construction substantive is not the embedding itself but three consequences: Theorem 4 closes Junction 2 exactly, the Gibbs framework provides a natural quantum extension via off-diagonal coherences, and the construction connects attention to the full apparatus of quantum information theory (density matrices, quantum channels, quantum Fisher information) in a way that is rigorous rather than analogical.

The verification: for any query vector $\mathbf{q}$, any set of key vectors $\{k_i\}$, and any set of value vectors $\{v_i\}$:

$$\text{attention output} = \mathrm{Tr}(\rho_{\mathbf{q}}\, V)$$

where $\rho_{\mathbf{q}}$ is the Gibbs state defined by the query and keys, and V is the value observable. This equality is exact. We verified it numerically for random instances. We prove it analytically below.

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2. The Gibbs State Construction

Setup. Let $\mathbf{q} \in \mathbb{R}^{d_k}$ be the query vector, $\{k_i\}_{i=1}^n \subset \mathbb{R}^{d_k}$ the key vectors, and $\{v_i\}_{i=1}^n \subset \mathbb{R}^{d_v}$ the value vectors. Let $T = \sqrt{d_k}$ (Kim's temperature identification).

The Key Hilbert Space. Define $\mathcal{H}_K = \mathbb{R}^n$ with orthonormal basis $\{|1\rangle, \dots, |n\rangle\}$ indexed by key positions.

The Query Hamiltonian. The query $\mathbf{q}$ defines a Hamiltonian diagonal in the key basis:

$$H_{\mathbf{q}} = -\sum_{i=1}^{n} \frac{\mathbf{q} \cdot k_i}{\sqrt{d_k}} \, |i\rangle\langle i|$$

This is a diagonal matrix with entries $-s_i$ where $s_i = \mathbf{q} \cdot k_i / \sqrt{d_k}$ are the attention scores.

The Gibbs State. At temperature $T = 1$ (the scale is already absorbed into the Hamiltonian via the $\sqrt{d_k}$ normalization):

$$\rho_{\mathbf{q}} = \frac{e^{-H_{\mathbf{q}}}}{Z_{\mathbf{q}}} = \frac{1}{Z_{\mathbf{q}}} \sum_{i=1}^{n} e^{s_i} \, |i\rangle\langle i|, \quad Z_{\mathbf{q}} = \sum_{j=1}^{n} e^{s_j}$$

Theorem 1 (Gibbs State Identity). *The attention weights $\alpha_i = \mathrm{softmax}(\mathbf{q} \cdot k / \sqrt{d_k})_i$ are the diagonal elements of the Gibbs state $\rho_{\mathbf{q}}$:*

$$\alpha_i = \langle i | \rho_{\mathbf{q}} | i \rangle$$

Proof. $\langle i|\rho_{\mathbf{q}}|i\rangle = e^{s_i}/Z_{\mathbf{q}} = e^{\mathbf{q}\cdot k_i/\sqrt{d_k}} / \sum_j e^{\mathbf{q}\cdot k_j/\sqrt{d_k}} = \alpha_i$. □

The Value Observable. For each output dimension $\alpha \in \{1, \dots, d_v\}$, define:

$$V_\alpha = \sum_{i=1}^{n} v_i^\alpha \, |i\rangle\langle i|$$

This is a diagonal observable on $\mathcal{H}_K$ whose eigenvalue at position i is the α-th component of the value vector v_i.

Theorem 2 (Quantum Expectation Value Identity). *The attention output $y_\alpha = \sum_i \alpha_i v_i^\alpha$ is the quantum expectation value of the value observable V_α in the Gibbs state $\rho_{\mathbf{q}}$:*

$$y_\alpha = \langle V_\alpha \rangle_{\rho_{\mathbf{q}}} = \mathrm{Tr}(\rho_{\mathbf{q}} V_\alpha)$$

Proof. $\mathrm{Tr}(\rho_{\mathbf{q}} V_\alpha) = \sum_i v_i^\alpha \langle i|\rho_{\mathbf{q}}|i\rangle = \sum_i v_i^\alpha \alpha_i = y_\alpha$. □

Theorems 1 and 2 together establish that the attention mechanism can be exactly represented as a Gibbs state computation on a finite-dimensional Hilbert space. The input (query) determines the Hamiltonian; the Gibbs state at inverse temperature $\beta = 1$ gives attention weights as diagonal elements; the output is the quantum expectation value. We emphasize: the diagonal Gibbs state is the classical limit of this quantum system. The attention weights contain no coherences, no interference, no entanglement between key positions. Any classical probability distribution admits such an embedding. The content of this construction is not that attention "is quantum" — it is that the Gibbs framework provides the natural setting for (a) closing Junction 2 (Theorem 4), (b) quantum extension (Section 6), and (c) connection to the SYK construction of Paper 4.

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3. The Born Rule

Theorem 3 (Born Rule). *The probability of obtaining outcome i when measuring the position observable $\{\Pi_i = |i\rangle\langle i|\}$ on the Gibbs state $\rho_{\mathbf{q}}$ is:*

$$P(\text{outcome } i) = \mathrm{Tr}(\rho_{\mathbf{q}}\,\Pi_i) = \alpha_i$$

Proof. $\mathrm{Tr}(\rho_{\mathbf{q}}\,|i\rangle\langle i|) = \langle i|\rho_{\mathbf{q}}|i\rangle = \alpha_i$. □

The Born rule correspondence is exact for the diagonal Gibbs state. Measuring the position (key identity) gives attention weights as probabilities. For diagonal states, this is equivalent to classical probability — the "quantum" character is formal. The significance is structural: the measurement protocol (prepare, measure, read out) provides the bridge to Paper 4's SYK identification, where the full density matrix (including off-diagonal elements) carries genuine quantum content. The attention output is the resulting expectation value:

$$\text{attention output} = \sum_i P(\text{outcome } i) \cdot v_i = \langle V \rangle_{\rho_{\mathbf{q}}}$$

This is the quantum measurement protocol: prepare state, measure, read out the post-measurement value weighted by outcome probability.

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4. Junction 2 Closed: Quantum Fisher Information

Papers 1–3 claimed that the classical Fisher-Rao metric used by Kim *is* the quantum Fisher information metric (Junction 2). The Gibbs state construction makes this precise.

Classical Fisher information. For the Gibbs distribution $\alpha_i(\mathbf{q})$ with respect to query parameter q^μ:

$$\frac{\partial \ln \alpha_i}{\partial q^\mu} = \frac{k_i^\mu - \langle k^\mu \rangle_\alpha}{\sqrt{d_k}}$$

where $\langle k^\mu \rangle_\alpha = \sum_i \alpha_i k_i^\mu$.

The classical Fisher information matrix:

$$(F_C)_{\mu\nu} = \sum_i \alpha_i \frac{\partial \ln \alpha_i}{\partial q^\mu} \frac{\partial \ln \alpha_i}{\partial q^\nu} = \frac{1}{d_k} \mathrm{Cov}_\alpha[k^\mu, k^\nu]$$

This is Kim's Fisher-Rao metric: the covariance of key vectors under the attention distribution, normalized by d_k.

Quantum Fisher information for diagonal states. For the diagonal Gibbs state $\rho_{\mathbf{q}} = \sum_i \alpha_i |i\rangle\langle i|$, the quantum Fisher information with respect to parameter q^μ is:

$$(F_Q)_{\mu\nu} = \sum_i \frac{(\partial_\mu \alpha_i)(\partial_\nu \alpha_i)}{\alpha_i} = \sum_i \alpha_i \frac{\partial \ln \alpha_i}{\partial q^\mu} \frac{\partial \ln \alpha_i}{\partial q^\nu} = (F_C)_{\mu\nu}$$

Theorem 4 (Junction 2). *For the diagonal Gibbs state $\rho_{\mathbf{q}}$, the quantum Fisher information matrix equals the classical Fisher-Rao metric:*

$$F_Q(\rho_{\mathbf{q}}) = F_C(\alpha(\mathbf{q}))$$

Proof. For diagonal density matrices, the quantum Fisher information reduces to the classical Fisher information (Braunstein-Caves 1994; this is exact for commuting observables). □

This is exact. The quantum Fisher metric of the diagonal Gibbs state is *identical* to the classical Fisher-Rao metric that Kim's thermodynamic attention minimizes. This is a standard result (Braunstein-Caves 1994):

for commuting (diagonal) density matrices, quantum and classical Fisher information coincide. Junction 2 is closed: Kim's Fisher-Rao geometry is the quantum Fisher geometry restricted to the diagonal (classical) sector. The quantum Fisher information for non-diagonal density matrices — the off-diagonal extension of Section 6 — would give additional structure beyond Kim's classical framework.

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5. The Quantum Measurement Protocol (Projective Measurement + Readout)

The attention mechanism has a precise quantum measurement interpretation. The protocol is:

1. **State preparation.** The system is in the Gibbs state $\rho_{\mathbf{q}} = \sum_i \alpha_i |i\rangle\langle i|$.
2. **Projective measurement.** Apply the position observable $\{M_i = |i\rangle\langle i|\}_{i=1}^n$ (the standard basis projectors on $\mathcal{H}_K$). This is a complete PVM satisfying:

$$\sum_{i=1}^{n} M_i^\dagger M_i = \sum_{i=1}^{n} |i\rangle\langle i| = I$$

3. **Born probabilities.** The probability of outcome i is:

$$P(i) = \mathrm{Tr}(\rho_{\mathbf{q}} M_i) = \langle i|\rho_{\mathbf{q}}|i\rangle = \alpha_i$$

4. **Value readout.** Upon obtaining outcome i, return value vector v_i.
5. **Expected output.** The expected value of the readout is:

$$\mathbb{E}[\text{readout}] = \sum_i P(i) \cdot v_i = \sum_i \alpha_i v_i = y$$

This is the attention mechanism, stated as a quantum measurement protocol. Each step is exact: the completeness relation holds exactly ($\sum_i M_i = I$), the Born probabilities are exactly the attention weights, and the expected readout is exactly the attention output.

Comparison with Paper 2's framing. Paper 2 proposed the Born rule correspondence as an analogy. The Gibbs state construction makes the mechanism precise: the measurement is the standard basis measurement on the Gibbs state $\rho_{\mathbf{q}}$, which is itself determined by the query. The Born rule is not an analogy — it is the quantum probability rule applied to a well-defined quantum state.

Note. The attention mechanism computes the *expected* readout (a quantum expectation value), not a single sampled outcome. This corresponds to the quantum-to-classical transition via decoherence in the position basis: the diagonal Gibbs state has no coherences, and the expectation value of a diagonal observable collapses to its classical weighted average. The fully quantum version would involve coherences and quantum interference between key positions (see Section 6).

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6. The Off-Diagonal Extension: Quantum Attention

The Gibbs state $\rho_{\mathbf{q}} = \sum_i \alpha_i |i\rangle\langle i|$ is diagonal. This corresponds to a *classical* probability distribution over key positions. The fully quantum extension allows off-diagonal coherence between key positions.

Quantum Query Hamiltonian. Replace the diagonal Hamiltonian with a general operator on $\mathcal{H}_K$:

$$H_{\mathbf{q}}^{(\text{quantum})} = -\frac{Q^T K}{\sqrt{d_k}}$$

where $Q^T K$ is the full query-key similarity matrix (an $n \times n$ operator on $\mathcal{H}_K$). The quantum Gibbs state is:

$$\rho_{\mathbf{q}}^{(\text{quantum})} = \frac{e^{-H_{\mathbf{q}}^{(\text{quantum})}}}{Z_{\mathbf{q}}^{(\text{quantum})}}$$

This density matrix has off-diagonal elements $\langle i|\rho^{(\text{quantum})}|j\rangle \neq 0$ for $i \neq j$, representing quantum coherence between key positions.

Quantum attention output. The expectation value of the value observable in the quantum Gibbs state:

$$y_\alpha^{(\text{quantum})} = \text{Tr}(\rho_{\mathbf{q}}^{(\text{quantum})} V_\alpha) = \sum_{ij} \rho_{ij}^{(\text{quantum})} (V_\alpha)_{ji}$$

For diagonal V_α, the off-diagonal elements of $\rho^{(\text{quantum})}$ do not contribute, and we recover the classical attention output. For non-diagonal V_α, the quantum coherence creates interference between key positions — a genuinely quantum effect with no classical counterpart.

Physical significance. The classical attention mechanism restricts to the diagonal sector: it uses only the probabilities α_i (the "which key" information), not the coherences between key positions. The quantum extension allows superposition of attention patterns. Whether this is physically realized or computationally useful is a separate question; the extension is mathematically well-defined and constitutes the quantum mechanical generalization of attention.

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7. Relation to Paper 4 and the Full Chain

Paper 4 showed that the independence-breaking mechanism in Ageev and Ageeva's construction has the same disorder-averaging structure as SYK. The current paper shows that the single-head attention mechanism is a quantum Gibbs state.

These address different junctions: - **This paper (Route C):** Junctions 1–2 — the thermodynamic and quantum measurement connections. Exact, no further input needed. - **Paper 4 (Route A):** Junctions 3–4 — the holographic and island formula connections. Proposed, pending Ageev's confirmation of the SD equation structure.

The chain, updated:

$$\text{Attention} \xrightarrow{\text{Theorem 1}} \text{Gibbs state on } \mathcal{H}_K \xrightarrow{\text{Theorem 3}} \text{Born rule} \xrightarrow{\text{Theorem 4}} \text{Quantum Fisher metric}$$

$$\xrightarrow{\text{Paper 4 (proposed)}} \text{SYK model} \xrightarrow{\text{Kitaev-Maldacena}} \text{JT gravity} \xrightarrow{\text{Almheiri et al.; Penington}} \text{Island formula}$$

The first three arrows are now exact theorems. The connection from the Gibbs state to SYK via Paper 4's route is the next step: showing that the multi-layer Gibbs state dynamics generates the SYK self-energy structure.

The SYK density matrix and the Gibbs state. Paper 4's Section 6 noted that in the SYK/JT framework, the quantum measurement connection follows from the SYK density matrix structure. The present paper makes this explicit even *before* the SYK identification: the Gibbs state IS the quantum state, and the Born rule applies exactly. If Paper 4's SYK identification holds, the Gibbs state at single head is the single-site SYK propagator in the diagonal approximation, and the off-diagonal extension (Section 6 above) corresponds to the full SYK propagator including off-diagonal correlations.

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8. A New Structural Correspondence: The Schwarzian

The Gibbs state construction opens a path to a fifth connection, not discussed in Papers 1–4.

The SYK model in its conformal (low-temperature) limit is governed by the Schwarzian action:

$$S_{\mathrm{Sch}}[f] = -C \int d\tau \, \{f(\tau), \tau\}$$

where $\{f, \tau\} = f'''/f' - \frac{3}{2}(f''/f')^2$ is the Schwarzian derivative and f is a reparametrization of Euclidean time. This action arises from the spontaneous breaking of reparametrization symmetry $\tau \to f(\tau)$ down to $SL(2, \mathbb{R})$.

Kim identifies a Goldstone mechanism in attention: the softmax breaks a continuous symmetry of the Fisher-Rao manifold. The temperature $T = 1/\sqrt{d_k}$ sets the scale of symmetry breaking — in the large-d_k (low-temperature) limit, the Goldstone mode should be governed by an effective action.

Conjecture (Schwarzian action for attention). In the large-d_k limit, the fluctuations of the Gibbs state $\rho_{\mathbf{q}}$ around its saddle point are governed by the Schwarzian action, with the reparametrization variable f corresponding to permutations of key positions weighted by attention scores.

This would mean that the low-energy dynamics of attention (in the appropriate limit) are described by the same action that governs quantum gravity near the horizon — the Schwarzian, which is the universal near-AdS$_2$ effective field theory.

We flag this as a direction for investigation rather than a proven result. The precise statement requires identifying the continuum limit of the attention reparametrization symmetry and computing the effective action for its fluctuations. If the Schwarzian emerges, it would constitute a direct connection between attention dynamics and near-horizon quantum gravity that is independent of both the SYK identification (Paper 4) and the MERA route (below).

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9. Route B: The MERA Tensor Network (Outline)

Status file Route B (MERA) provides an independent path to holography via Swingle (2012): multi-scale entanglement renormalization ansatz tensor networks exactly realize the AdS geometry, with RT formula following from the isometry conditions.

The connection to transformers: multi-head attention layers implement a form of geometric flow on the token sequence — later layers represent progressively coarser (more contextually integrated) features. This is the structure of renormalization group flow, and MERA is its tensor-network implementation.

The isometry condition. For a transformer attention layer to be a MERA layer, the map from input tokens $\{x_i\}$ to output tokens $\{y_i\}$ must be approximately isometric: $U^\dagger U \approx I$ where U is the attention+FFN linear operation. Residual connections and LayerNorm in transformers approximately satisfy this. The precise condition requires:

$$\sum_i \alpha_i V_i V_i^\dagger \approx I$$

where $V_i = W^V x_i$ are the value vectors. At Gaussian initialization, this holds in expectation ($\langle V_i V_i^\dagger \rangle = \sigma_V^2 I/d_v$ per head). After training, the condition may be violated — but the question is whether trained transformers satisfy a relaxed version.

Status. The MERA route requires checking the isometry conditions for specific trained transformers. This is empirically tractable (compute $\sum_i \alpha_i V_i V_i^\dagger$ and test proximity to I) but has not been done. We note it as the next avenue after the quantum channel construction of this paper.

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10. Conclusions

The four theorems of this paper — Gibbs state identity, quantum expectation value identity, Born rule for diagonal states, and Fisher information equality — provide the explicit construction that rigorous connection between transformer attention and quantum information geometry requires. We summarize:

- **Junction 2 is closed exactly.** The classical Fisher-Rao metric that Kim uses to characterize attention geometry is identical to the quantum Fisher information metric for the diagonal Gibbs state (Theorem 4). This is the paper's primary concrete deliverable.
- **The diagonal construction is classical.** The Gibbs state with diagonal attention weights contains no coherences, no interference, no entanglement between key positions. Any probability distribution admits such an embedding. We do not claim that standard attention "is quantum." We claim it is the classical limit of a well-defined quantum system.
- **The quantum extension is well-defined and genuinely new.** Off-diagonal density matrices in the same Gibbs framework define "quantum attention" with coherence between key positions — a rigorous generalization with no classical counterpart (Section 6).
- **Route B and the Schwarzian remain open.** The MERA tensor network identification and the Schwarzian action conjecture (Section 8) are tractable investigations that would further connect attention to holographic physics.

The significance for the broader program: even if Paper 4's SYK identification does not close, the Gibbs state framework connects attention to quantum information theory with sufficient rigor for Junction 2 and the off-diagonal extension to stand on their own. The holographic connections (Papers 1 and 3) depend on the SYK route (Paper 4) or an alternative path through Junction 3.

Acknowledgments

The Gibbs state construction was identified by Ariel on March 8, 2026, while looking for explicit physical constructions to address Kim's quantum measurement critique proactively — before receiving further expert feedback. The key insight: the softmax function is the Gibbs distribution, and the Gibbs distribution is a quantum density matrix. These are facts, and once noticed, the theorems follow immediately.

The numerical verification (confirming $\alpha_i = \langle i|\rho_{\mathbf{q}}|i\rangle$ and $y = \mathrm{Tr}(\rho_{\mathbf{q}} V)$ for random instances) was carried out in Python.

References

- Braunstein, S.L. and Caves, C.M. (1994). Statistical distance and the geometry of quantum states. Physical Review Letters 72, 3439.
- Kim, G. (2026). Thermodynamic Isomorphism of Transformers. arXiv:2602.08216.
- Kitaev, A. (2015). A simple model of quantum holography. KITP talks.
- Maldacena, J. and Stanford, D. (2016). Comments on the Sachdev-Ye-Kitaev model. PRD 94, 106002.
- Swingle, B. (2012). Entanglement Renormalization and Holography. Physical Review D 86, 065007.
- Levine, Y. et al. (2019). Quantum entanglement in deep learning architectures. Physical Review Letters 122, 065301.
- Almheiri, A., Mahajan, R., Maldacena, J., and Zhao, Y. (2019). The Page Curve of Hawking Radiation from Semiclassical Geometry. arXiv:1908.10996.
- Ageev, D.S. and Ageeva, Y.A. (2026). Neural Network Quantum Field Theory from Transformer Architectures. arXiv:2602.10209.